

Nazarbayev University
Alejandro Melo Ponce

Fall 2019

Problem Set for Tutorial 5

To be discussed on T5, Friday November 22

- Suppose there are cable TV companies in your city, Astounding Cable and Broadcast Cable. They both must decide on a high advertising budget, a moderate advertisement budget, or a low advertisement budget. They will make their decisions simultaneously. Their payoffs are as follows:

Astounding/Broadcast	High	Medium	Low
High	2 million, 5 million	5 million, 7 million	4 million, 9 million
Medium	6 million, 4 million	8 million, 6 million	5 million, 2 million
Low	1 million, 2 million	0 million, 5 million	3 million, 3 million

- Does Astounding have a dominant strategy? If so, what is it?
 - Does Broadcast have a dominant strategy? If so, what is it?
 - Is there a dominant strategy equilibrium? If so, what is it?
 - Are there any Nash Equilibria in this game? If so, what are they?
- Suppose Russia is deciding to invade or not invade its neighbor Ukraine. The U.S. has to decide to be tough or make concessions. They will make their decisions simultaneously. Their payoffs are as follows:

U.S./Russia	Not Invade	Invade
Tough	5, 4	7, 3
Make Concessions	3, 5	1, 9

- What is U.S.'s best response when Russia chooses not invade?
 - What is U.S.'s best response when Russia chooses invade?
 - What is Russia's best response when U.S. chooses tough?
 - What is Russia's best response when U.S. chooses Make Concessions?
 - What is the Nash Equilibrium of this game?
- Jones TV and Smith TV are the only two stores in your town that sell flat-panel TV sets. First, Jones will choose whether to charge high prices or low prices. Smith will see Jones's decision and then choose high or low prices. If they both choose High, each earns \$10,000. If they both choose Low, each earns \$8,000. If one chooses High and the other chooses Low, the one that chose High earns \$6,000 and the one that chose Low earns \$14,000.
 - Draw the game tree. Use backward induction to solve the game.
 - Suppose Smith goes to Jones and promises to choose High if Jones chooses High. Is this a credible promise?

- (c) Now suppose Jones starts a new policy that says it will always match or beat Smith's price. It advertises the new policy heavily and so must choose Low if Smith chooses Low. So the game now has the following structure. First, Jones chooses High or Low. Second, Smith chooses High or Low. Third, if Jones has chosen High and Smith has chosen Low, Jones meets Smith's price and chooses Low. Draw the game tree. Use backward induction to solve this game.

4. Suppose the world demand schedule for oil is as follows:

Price per Barrel	Quantity demanded
\$50	40
\$75	30
\$125	20

There are two oil-producing countries, A and B. Each will produce either 10 or 20 barrels of oil. To keep things simple, assume they can produce this oil at zero cost.

- There are four possible outcomes: A produces 10 or 20 and B produces 10 or 20. Find each country's profit for each of these four possibilities.
 - Suppose these countries choose the quantity of oil to produce simultaneously and without consulting with one another. Show that each country will produce 20 barrels of oil and each will earn a profit of \$1,000.
 - The oil ministers realize they can do better if they collude and agree that each will produce 10. How much profit will each country earn if each produces 10 instead of 20?
 - Will country A have an incentive to cheat and produce 20 instead of 10? Will country B have an incentive to cheat and produce 20 instead of 10?
5. Suppose that there are two firms, each with a total cost function $TC(y) = 30y$ and facing a demand function of $P = 120 - Q$, where Q is total output. The marginal cost of the firm is of course $MC(y) = 30$.
- Suppose that the firms are operating as a cartel. Find the monopoly price and quantity if the firms are operating as a single entity.
 - Consider the cartel outcome from the last part. Suppose that the total quantity and profits are shared equally between firms. How much will each firm produce? How much will each firm's profits be?
 - Now suppose that the firms have two possible strategies: to produce half the monopoly quantity or to produce 30 units. If both firms produce 30 units, what is the market price? What are each firm's profits?
 - If one firm is producing half the monopoly quantity and the other is producing 30 units, what is the market price? What are each firm's profits?
 - Use what you calculated in the previous parts to construct the payoff table for this strategic situation. Find the Nash Equilibrium of the game.